1st Place: Aspect 7 – Simplify or decompose a situation or problem to make it amenable to mathematical analysis

- **Analysis**: This is the most central competency of the question. The problem explicitly provides two methods: a tedious one ("decompose-calculate-sum") and a more efficient one ("only 4 length measurements needed"). Students need to understand that the essence of the efficient method is to greatly simplify the complex planar figure (the apartment floor plan) by viewing it as a more regular composite shape (e.g., a large rectangle minus several smaller rectangles, or composed of several rectangles). This simplifies the area calculation to depend on only a few key dimensions.
- **Manifestation in the Question**: Students must simplify the floor plan, which contains multiple rooms, corridors, and irregular shapes, into a geometric model describable by a small number of lengths. This requires students to possess strong "simplification" thinking to see the structure behind the figure.

2nd Place: Aspect 5 – Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Analysis**: To find those "4 lengths," students must identify the hidden regular mathematical structure within the floor plan.
- **Manifestation in the Question**: Apartment floor plans are typically composed of rectangular or nearly rectangular rooms arranged in an aligned manner. Efficient methods are often based on this structure: the overall outline of the apartment can be approximated as a **large rectangle**, whose area is determined by its length and width (2 lengths). Then, internal missing parts (e.g., recessed corners, courtyards) or protruding parts (e.g., balconies) can be sized by measuring 1-2 additional lengths, allowing for addition or subtraction from the large rectangle's area. Students need to recognize this generic pattern of "bounding rectangle plus/minus adjustments."

3rd Place: Aspect 6 – Identify and describe the mathematical aspects of a real-world problem, including identifying significant variables

- **Analysis**: After simplifying the model and identifying the structure, the next step is to determine which measurements are crucial, i.e., identifying the "variables" in the model.
- **Manifestation in the Question**: Students need to clearly specify which four lengths. These typically include: the **total length (L)** and **total width (W)** of the entire apartment, and **1-2 key internal dimensions** used to correct for internal recesses or external protrusions (e.g., the depth `d` of a recess, the width `w` of a protrusion). Correctly identifying these 4 variables is key to successful modeling.

4th Place: Aspect 1 – Select a mathematical description or representation that describes a problem

- **Analysis**: Students need to choose a mathematical way to describe this efficient area estimation method.
- **Manifestation in the Question**: They need to select a specific mathematical description such as **"area of a bounding rectangle plus/minus correction areas,"** rather than remaining at the vague statement "measure 4 lengths." When marking the lengths, they are essentially using line segments—a geometric representation—to define this model.

5th Place: Aspect 4 – Read, decode, and make sense of images to create a model of the situation

- **Analysis**: This is the starting point for all work. Students must correctly interpret the visual information of the apartment floor plan, understand the layout of walls, rooms, and terrace, and view it as a measurable geometric object.
- **Manifestation in the Question**: Students need to comprehend the floor plan, understand that lines represent walls, blank areas represent rooms or the terrace, and visualize the spatial relationships of these parts.

6th Place: Aspect 10/11 – Use mathematical tools (e.g., diagrams, markings) to describe mathematical structures and relationships

- **Analysis**: The task "Mark on the plan" directly requires students to use graphical tools (making marks on the diagram) to concretize their mathematical model.
- **Manifestation in the Question**: By drawing four key line segments on the floor plan, students are using intuitive geometric language to describe the mathematical model and key variables they have identified, visualizing abstract thinking.

7th Place: Aspect 8 – Recognise aspects of a problem that correspond with known problems or concepts

- **Analysis**: Students might relate this problem to known topics they have learned, such as "area calculation of composite figures" or "finding the area of irregular shapes using decomposition or complement methods."
- **Manifestation in the Question**: Realizing that the "measure 4 lengths" method likely corresponds to known problem-solving strategies like "completing an irregular figure into a rectangle and subtracting extra parts" or "dividing into several rectangles and summing their areas."

8th Place: Aspect 12 – Identify the constraints, assumptions, and simplifications in a mathematical model

- **Analysis**: Any efficient model is a simplification of reality and necessarily involves assumptions.
- **Manifestation in the Question**: The method assumes wall thickness is negligible or that the terrace can be treated equally with indoor area ("including the terrace"). It assumes the floor plan is regular and that minor irregularities have little impact on the total area estimate. This is an **estimation** model, not an exact calculation. Understanding this is part of complete mathematical thinking.

第1位：方面 7 - 简化或分解情境，使其易于进行数学分析

- **分析**：这是本题最核心的素养。题目明确提供了两种方法：一种是繁琐的“分解-计算-求和”，另一种是更高效的“仅需测量4个长度”。学生需要理解，高效方法的本质是对复杂平面图形（公寓平面图）进行极大的简化，将其视为一个更规则的复合图形（例如，一个大矩形减去几个小矩形，或由几个矩形组合而成），从而将面积计算简化为仅依赖少数几个关键尺寸。
- **题目体现**：学生必须将包含多个房间、走廊、不规则形状的平面图，简化为一个能用少数长度描述的几何模型。这要求学生具备强大的“简化”思维，看到图形背后的结构。

第2位：方面 5 - 在问题或情境中识别数学结构（规律、关系、模式）

- **分析**：为了找到那“4个长度”，学生必须在平面图中识别出隐藏的规律性数学结构。
- **题目体现**：公寓平面图通常由矩形或近似矩形的房间组成，这些房间以对齐的方式排列。高效方法往往基于这样的结构：公寓的总轮廓可以近似为一个**大矩形**，其面积由长和宽（2个长度）决定。然后，内部缺失的部分（如凹角、天井）或突出的部分（如阳台）可以通过再测量1-2个长度来确定其大小，从而从大矩形面积中加上或减去相应部分。学生需要识别出这种“外包矩形加减调整”的通用模式。

第3位：方面 6 - 识别并描述现实问题中的数学方面，包括识别重要变量

- **分析**：在简化模型和识别结构之后，下一步就是确定哪些测量值是关键的，即识别模型中的“变量”。
- **题目体现**：学生需要明确指出是哪四个长度。这通常包括：整个公寓的**总长度（L）**和**总宽度（W）**，以及用于修正内部凹陷或外部凸起部分的**1-2个关键内部尺寸**（例如，凹陷部分的深度d、凸出部分的宽度w）。正确识别这4个变量是成功建模的关键。

第4位：方面 1 - 选择描述问题的数学描述或表征

- **分析**：学生需要选择用什么样的数学方式来描述这个高效的面积估算方法。
- **题目体现**：他们需要选择 **“用外接矩形面积加减修正部分面积”** 或类似的具体数学描述，而非停留在“测量4个长度”的模糊陈述。在标记长度时，他们实际上是在用线段这种几何表征来定义这个模型。

第5位：方面 4 - 阅读、解码并理解图像，以建立情境模型

- **分析**：这是所有工作的起点。学生必须正确解读公寓平面图这一视觉信息，理解墙壁、房间、 terrace 的布局，将其视为一个可测量的几何对象。
- **题目体现**：学生需要看懂平面图，理解图中的线条代表墙壁，空白区域代表房间或 terrace，并能想象这些部分的实际空间关系。

第6位：方面 10/11 - 使用数学工具（如图形、标记）描述问题中的数学结构与关系

- **分析**：题目的任务“Mark on the plan”直接要求学生使用图形工具（在图上做标记）来具体化他们的数学模型。
- **题目体现**：通过在平面图上画出四条关键的线段，学生就是在用直观的几何语言描述他们识别出的数学模型和关键变量，将抽象思维可视化。

第7位：方面 8 - 识别与已知问题或数学概念的对应

- **分析**：学生可能将此问题与他们学过的“组合图形面积计算”或“用分割法/填补法求不规则图形面积”等已知问题联系起来。
- **题目体现**：意识到“测量4个长度”的方法很可能对应着“将不规则图形补成一个矩形，再减去多余部分”或“分割成几个矩形求和”的已知解题策略。

第8位：方面 12 - 识别数学模型中的约束、假设与简化

- **分析**：任何高效的模型都是对现实的简化，必然包含假设。
- **题目体现**：该方法假设墙壁厚度可以忽略不计，或者 terrace 与室内面积可以同等对待（“包括 terrace”）。它假设平面图是规则的，轻微的凹凸不规则性对总面积估算影响不大。这是一个**估算**模型，而非精确计算。理解这一点是完整数学思维的一部分。